Dazzling colours are exhibited by thin films of oil spread on the surface of water and also by soap bubbles as shown in the figure. These colours are due to interference of white light undergoing multiple reflections from the top and the bottom surfaces of thin films. The colour depends upon the thickness of the film, refractive index of the film and also the angle of incidence of the light.



$$\frac{1}{n}=\frac{1}{3} \text{ (or) } n=3$$

Thus, the 3rd bright fringe of λ_1 and 4th bright fringe of λ_2 coincide at the least distance y.

The least distance from the central fringe where the bright fringes of the two wavelengths coincides is, $y_n = n\frac{\lambda D}{d}$

$$y_n = 3\times\frac{560\times10^{-9}\times1}{3\times10^{-3}} = 560\times10^{-6}\,\text{m}$$

$$y_n = 0.560\times10^{-3}\ \text{m} = 0.560\ \text{mm}$$

7.3.6 Interference in thin films

Let us consider a thin film of transparent material of refractive index μ (here refractive index is not represented as *n*, not to confuse with order of fringe *n*) and thickness *d*. A parallel beam of light is incident on the film at an angle *i* as shown in Figure 7.16. The wave is divided into two parts at the point of incidence, as reflected and refracted lights. The refracted part, which enters into the film, again gets divided at the lower surface into two parts; one is transmitted out of the film and the other is reflected back into the film. The reflected as well as refracted parts are further formed as multiple reflections take place inside the film. The interference occurs in both the reflected and transmitted light.

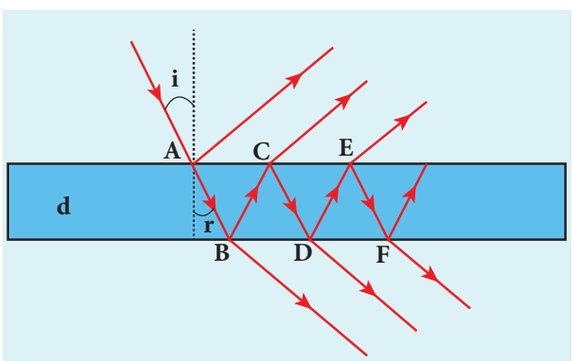


Figure 7.16 Interference in thin films

For transmitted light

The light transmitted may interfere to produce a resultant intensity. Let us consider the path difference between the two light waves transmitted from *B* and *D*. The two waves moved together and remained in phase

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